

JAVA

FULL STACK + REACT JS & GEN AI INTEGRATION



React JS is the most popular JavaScript framework and platform for developing client-side (front-end) mobile and desktop web apps or single page applications (SPAs).

HTML

INTRODUCTION TO HTML

- headings
- paragraph
- formatting tags
- image
- anchor
- lists
- table
- ID
- class
- iframe
- div
- span
- form
- audio
- video
- html vs html5

CSS

Introduction to CSS

ways to use CSS (inline, internal, external)

css selectors

css color

border

padding

margin

height

width

lists

text

display (none, inline, block, inline-block)

flex

grid

position

z-index

media

Java Script

Js Basics

- keywords
- identifiers
- datatypes
- variables and constants
- tokens
- operators

Decision Making Statements

Conditions Statements Or Selection Statements

- if
- if else
- nested if else
- if else if ladder
- switch case

Iteration Statements Or Loops

- for
- while
- do while
- for in
- for of

Jump Statements

- continue
- break

Functions

- function declarations, definitions and calling
- types of functions
- anonymous functions
- arrow functions or arrow functions
- self invoked function (IIFE)
- storage classification or variable scope (var let const and without keywords)
- recursion

Strings Methods

array Methods

Global Execution Context and Function execution context

- scope chain rule and lexical scoping
- closure

Objects

- literal and keys
- object functions
- nested objects
- Object Prototypes
- setter and getter methods

Set and Map

WeakMap and WeakSet

Dates and Time

OOPS

- literal and keys
- object functions

- nested objects
- Object Prototypes
- setter and getter methods

DOM Manipulation

- DOM
- BOM
- event handling
- time based events

Advanced Js

How Javascript Works

hoisting

strict mode

closure

synchronous and asynchronous programming

Event propagation

bubbling

capturing

Event loop

function curring

ajax and json

calling api using ajax

callback hell

promise

async await

fetch

modules

version

Typescript

- Why Typescript
- Basic Types
- Class and Interfaces
- Modules

React Js

INTRODUCTION TO REACT JS

- What is React JS?
- What is SPA?
- DOM vs Virtual DOM
- Advantages and Disadvantages
- Key Features

ENVIRONMENTAL SETUP

- Node | NPM
- Installation of CLI
- Setup Project
- Directory Structure
- Code Editors
- How React JS Application Boot

BASIC FEATURES OF REACT JS

- React Concepts
- JSX and TSX

- Render Elements
- Function and Class Components
- Props and State
- Handling Events
- Dynamic Data Rendering
- Property Binding

KEY FEATURES OF REACT JS

- Conditional Rendering
- List and Keys
- Forms Handling
- Forms Validations

COMPONENT LIFECYCLE HOOK

- Understanding component lifecycle
- All Lifecycle Hooks

EVENT HANDLING REACT

- Understanding React Event System
- Passing arguments to event Handlers

NETWORK CALL

- Fetch
- Axios

CUSTOM SERVICES

- Introduction to Services
- Building a Service

LOCAL DATA STORAGE

- Local Storage
- Session Storage
- Cookies

ROUTING WITH REACT ROUTER

- Setting up React Router
- Configuring route with Route Component
- Making routes dynamic with Route Params
- Working with nested routes
- Link and NavLink
- Redirect Routes

UI COMPONENTS

- Material Design
- PrimeNG

INTRODUCTION TO REDUX

- Why Redux
- Install and setup
- Store, Reducer, actions
- Dispatcher
- High order Components
- mapStateToProps and map DispatchToProps usage

ADVANCE REDUX

- Async Actions
- Middleware
- Redux Thunk and Redux Saga

React Hooks

- Why We Need Hooks.
- Different Types Of Hooks

- Using State And Effect Hooks
- Usereducer, Useref Etc.
- Custom Hooks
- Rules Of Hooks

Third Party Modules

- Social Login
- Pagination
- Search
- Filter
- JWT Token
- File Upload
- Many More

Rest Js Testing

- Jest with Enzyme

Develop a CRUD Application in React Js

INTRODUCTION TO JAVA

- Why Java was Developed
- Application Areas of Java
- History of Java
- Platform Independency in Java
- USP of Java: Java Features
- Sun-Oracle Deal
- Different Java Platforms
- Difference between JDK,JRE,JVM
- Java Versions
- JVM Architecture
- Installing Java on Windows
- Understanding Path Variable: Why Set Path

CREATING FIRST JAVA PROGRAM

- Understanding Text Editors to Write Programs
- How to compile java file
- Byte Code and class file
- How to run class file

JAVA LANGUAGE FUNDAMENTALS

- Identifiers
- Keywords
- Variables
- Literals
- Data Types
- Operators
- Comments
- Looping Statements
- Condition Statements
- Type Casting

OOP IMPLEMENTATION (PIE)

- Why OOP
- OOP Concepts with Real life examples
- Class& it's Syntax
- Object& it's Syntax
- Reference Variable
- Constructors
- Instance(Non-Static)& Static Variables
- Instance(Non-Static) & Static Methods
- this Keyword and it's usages
- Object & Static Initializers(Anonymous Blocks)
- Understanding '+' Operator
- Inheritance& it's Syntax
- Types of Inheritance
- Object Class as Root of Java Class Hierarchy
- Variable Hiding
- Method Hiding
- Method Overriding
- Method Overloading
- Super keyword and it's usages
- Final keyword and it's usages
- Constructor Chaining
- Upcasting and Downcasting
- Static &Dynamic Binding
- Run Time Polymorphism
- Abstract Keyword(Abstract classes and methods)
- Understanding Interfaces
- Implementation of Encapsulation
- Association with Implementation

PACKAGES

- Understanding Packages
- Setting Class path
- Reading Input from Keyboard
- Access Modifiers
- With in Package & Outside Package Implements

NESTED TYPES

- Static Nested Class
- Non-static Nested Class
- Local Class
- Anonymous Class
- Nested Interface

ARRAYS

- General Definition of Array
- Advantages from Array
- Arrays in Java
- 1-d Arrays
- 2-d Arrays
- Jagged Arrays
- Array of reference type
- Operations on Arrays
- User Define Array & Object Type

COMMAND LINE ARGUMENTS AND WRAPPER CLASSES

- How to read command line arguments
- Wrapper Classes
- Parsing of Numeric Strings
- String representation of Primitives

EXCEPTION HANDLING

- Types of Runtime Errors
- Understanding Exceptions
- Exception Class Hierarchy
- Try & Catch Blocks
- Patterns of Catch Block
- Nested Try statements
- Throw, throws and finally
- Creating Custom Exceptions
- Checked & Unchecked Exceptions
- Assertion

WORKING WITH STRINGS

- What is String
- String Class
- Creating String Object
- Operations on String
- String Buffer Class and it's Methods
- Difference between String and StringBuffer class
- String Builder Class and it's Methods
- Difference between StringBuffer and StringBuilder

MULTITHREADED PROGRAMMING

- Multitasking: Why Concurrent Execution
- Multiprocessing v/s Multithreading
- Main Thread (Default Java Thread)
- Creating Child Threads and understanding context switching
- Thread States
- Thread Group
- Thread Synchronization: Methods and Blocks
- Inter-Thread communication
- Daemon Threads
- Deadlock

I/O STREAMS

- What is I/O
- Why Need Streams
- Byte Streams and Character Streams
- Read/Write operations with file
- Scanner Class
- Object Serialization& Deserialization
- Transient keyword
- File Class and it's Methods

SOCKET PROGRAMMING

- Understanding Fundamentals of a Network
- Socket and ServerSocket Classes
- InetAddress Class
- DatagramSocket and DatagramPacket Classes
- URL,URLConnection,HttpURLConnection Classes

REFLECTION

- Understanding the Need Of Reflection
- Getting information about class's modifiers, fields, methods, constructors and super classes
- Finding out constant and method declaration belong to an interface
- Creating an instance of the class whose name is not known until runtime
- Getting and setting values of an object's field if field name is unknown until runtime
- Invoking a method on an object if the method is unknown until runtime
- Invoking Private Methods

EXTENDED & UTILITY CONCEPTS

- Generics
- Lambda Expression
- Annotations
- Object Cloning
- Vargs
- Static-import
- Enum
- Static, Default and Private Methods of Interface
- Var Type
- Java Modules
- Stream API

COLLECTIONS FRAMEWORK

- What is Collection?
- What is Framework?
- Collections Framework
- Core Interfaces
- Collection, List, Queue,Deque
- Set,NavigableSet, SortedSet
- Map,NavigableMap, SortedMap
- Core Classes
- ArrayList, LinkedList,PriorityQueue,ArrayDeque
- HashSet,LinkedHashSet,TreeSet,
- HashMap,IdentityHashMap,WeakHashMap,LinkedHashMap,Tree Map
- Accessing a Collection via an Iterator
- Accessing List via ListIterator
- Accessing a Collection via for each loop
- Working with User Defined Objects
- The Comparator and Comparable Interfaces
- The Legacy classes and Interfaces.
- Enumeration, Vector ,Stack
- Hashtable, Properties

DATE & TIME API

- java.util.Date
- java.util.Calendar
- java.sql.Date

SYSTEM PROPERTIES & INTERNATIONALIZATION (I18N)

- Understanding Locale
- Resource Bundle
- Usage of properties file
- Fetching text from Resource Bundle
- Displaying the text in HINDI
- Displaying date in Hindi

INTRODUCTION TO SQL (PROJECT BASED)

DATABASE PROGRAMMING USING JDBC

- Need Of JDBC
- JDBC Drivers
- Statement, PreparedStatement, CallableStatement
- Scrollable and Updatable ResultSet
- Batch Updates
- Transaction
- Metadata
- Connection Data Base
- Oracle
- My SQL
- Mongo DB

JAVA EE(JAVA PLATFORM ENTERPRISE EDITION)

- Understanding the Concept of Java EE : JEE Specification
- Java EE Architecture
- Single Tier
- Two Tier
- Three Tier
- N-Tier
- Java EE Components
- Web Components
- Distributed(Business) Components
- Java EE Containers& Servers
- Web Container& Web Server(Apache Tomcat)
- EJB Container& Application Server(Weblogic,Glassfish,Websphere)
- Java EE Services
- JNDI Service
- Java Transaction Service
- JAAS

WEB PROGRAMMING

- Servlet
- JSP/JSTL
- Project Work

JAVA SERVER PAGES (JSP) & JSTL

- JSP Architecture
- JSP Elements
- JSP Objects
- Understanding JavaBeans
- Custom Tags
- Using tags of JSTL
- Expression Language

PROJECT CLASSES

- Front End Coding

- FORM DESIGNING
- HTML
- CSS
- JAVA SCRIPT
- BOOTSTRAP
- Back End Coding
- DATABASE DESIGNING
- Connecting forms to database
- Writing Business Logic
- Project Hosting

DESIGN PATTERN

- Why Design Patterns...?
- Front Controller
- Composite View
- Factory Pattern
- Singleton Pattern
- DAO Pattern

JAVA MAIL API

- Email System and Protocols
- Sending & Receiving Mails
- Handling Attachments

INTRODUCTION TO DISTRIBUTED PROGRAMMING

- RMI
- Web Services

INTRODUCTION TO RESTFULL SERVICES

- @PathParam
- @Path
- @FormParam
- @QueryParam
- @DefaultValue

OVERVIEW OF JPA FRAMEWORK

SPRING

- What is Spring?
- Spring modules
- Understanding dependency Injection
- Applying aspect-oriented programming

MAVEN DEPLOYMENT

- Maven Configuration
- Converting Maven to Eclipse
- Various Maven Command

BASIC BEAN WIRING

- Containing your Bean
- Creating bean
- Injecting into bean properties
- Auto wiring
- Controlling bean creation
- Aspect Oriented Programming

HITTING THE DATABASE

- Learning spring's data Access Philosophy
- Configuring a data source

- Using JDBC with Spring
- Working with JDBC Templates
- Using Spring's DAO Support Classes for JDBC
- Integrating Hibernate with Spring
- Caching

HIBERNATE

INTRODUCTION TO ORM

- Need of ORM
- Problems using JDBC Directly
- ORM Implementation

INTRODUCTION TO HIBERNATE

- Hibernate Architecture
- Hibernate configuration
- Hibernate's Support for Other Technologies
- Installing Hibernate
- A “Hello world” stand alone application

CREATING PERSISTING CLASSES

- Mapping a basic Java Class
- Mapping a Class with Binary Data
- Mapping a Serializable Class
- Mapping a class with Data/ calendar attributes
- Mapping a Read-only class
- Mapping a class using Versioning /Timestamps

MAPPING INHERITENCE WITH JAVA CLASSES

- Table-Per –class Hierarchy Mapping
- Table-Per –subclass Hierarchy Mapping
- Table-Per –concrete-subclass Hierarchy Mapping
- Persistence interfaces

WORKING WITH COLLECTIONS

- Associations
- Lazy initialization
- Mapping Maps/Sorted Maps
- Mapping Sets/Sorted Sets
- Mapping lists
- Mapping Arrays
- Mapping a Bidirectional Association

HIBERNATE CACHING

- How caching improves performance
- First level lache
- Second level cache

SPRING BOOT

- Introduction To Spring Boot
- Spring Boot & JDBCTemplate
- Spring Boot Annotation
- Spring Boot & JPA Hibernate
- Spring Boot Rest API
- Spring Boot MVC
- Spring Boot Security
- Introduction to Micro Services

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Generative AI Integration

Introduction to AI & Large Language Models

Understanding GPT Models

- What is OpenAI
- How to get API key
- Understanding Library
- How to generate text
- How to perform sentiment analysis
- How to generate image from text
- How to implement chatbot



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